

SinkKV: Sink-Protected Mixed-Precision KV Cache

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Abstract

This is a living paper shell for a preregistered Mac-first experiment. SinkKV tests whether protecting the first sink K/V positions while quantizing the rest of the KV cache recovers the quality loss of uniform FP4-style KV quantization. No positive result, GPU throughput, latency, HBM, or native FP4 claim is made yet.

1 Current Gate

The preregistered Mac gate compares BF16 K/V, uniform simulated MXFP4 K/V, and sink-protected simulated MXFP4 K/V. Promotion requires sink protection to recover at least half of the uniform low-precision quality gap on at least two model/length surfaces, with confidence intervals excluding the no-recovery boundary.

2 Claim Boundary

Allowed now: a preregistered method hypothesis and deterministic local simulation utilities. Not allowed: native FP4, vLLM integration, speed, memory, HBM, or benchmark-quality claims.